

TrustLot - Used Car Marketplace User Manual

Version 1.0

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1. Introduction

Welcome to **TrustLot**, a comprehensive full-stack used car buying and selling marketplace. This manual provides complete guidance for all users—buyers, sellers, and administrators—covering registration, navigation, feature usage, and troubleshooting.

Key Features:

- **Secure Authentication** with session-based access control
 - **Multi-role Support** for buyers, sellers, and administrators
 - **Complete Purchase Lifecycle** with admin and seller approval flows
 - **Wishlist Management** to save favorite cars
 - **Seller Ratings** for buyer confidence
 - **Support Ticket System** for issue resolution
 - **Admin Dashboard** with real-time analytics
-

2. System Overview

Architecture

TrustLot is built on a modern full-stack architecture:

- **Backend:** Spring Boot 3.3.5, Java 17
- **Database:** Apache Derby (file-based) or MySQL (configurable)
- **Authentication:** Custom session tokens (120-minute expiry)
- **API:** RESTful endpoints with JSON request/response
- **Base URL:** `http://localhost:8080` (development)

Core Concepts

Concept	Description
User	Can act as both buyer and seller simultaneously

Concept	Description
Admin	System administrator (only one allowed)
Car Listing	Requires admin approval before public visibility
Purchase Order	Two-step approval: Admin → Seller
Session Token	Authentication key for protected endpoints

3. Getting Started

3.1 Registration

Endpoint: POST /api/auth/register

Step 1: Fill Registration Form

```
{
  "username": "rahul1",
  "name": "Rahul Sharma",
  "phoneNumber": "9876543210",
  "email": "rahul@example.com",
  "password": "rahulsecure1234",
  "role": "USER"
}
```

Validation Rules:

- **Username:** Must be unique
- **Name:** Title case (e.g., “John Doe”)
- **Phone:** Indian 10-digit number starting with 6-9
- **Email:** Optional but recommended (must be unique if provided)
- **Password:** Minimum 10 characters (alphanumeric + special chars recommended)
- **Role:** Either USER or ADMIN

Important Notes:

- First registered ADMIN becomes the system administrator
- Only one ADMIN account allowed; subsequent ADMIN registrations are rejected
- Users can both buy and sell using a single USER account

Success Response:

```
{
  "message": "Registration successful. Please login to start a session."
}
```

3.2 Login

Endpoint: POST /api/auth/login

Step 1: Enter Credentials

```
{
  "username": "rahul1",
  "password": "rahulsecure1234"
}
```

Success Response:

```
{
  "message": "Login successful",
  "sessionToken": "abc123def456ghi789jkl",
  "role": "USER"
}
```

Step 2: Store Session Token

- Save the `sessionToken` in your application
- Include it as `X-Session-Token` header on all protected requests
- Sessions expire after 120 minutes of inactivity

Example Protected Request:

```
GET /api/cars/available
X-Session-Token: abc123def456ghi789jkl
```

3.3 Logout

Endpoint: POST /api/auth/logout

```
{
  "message": "Logout successful"
}
```

4. User Roles & Permissions

4.1 User (USER) Role

Dual Capability Model: A single USER account functions as both buyer and seller.

Permissions:

Action	Allowed
List cars for sale	Yes
Browse other cars	Yes
Purchase cars	Yes
Review sellers	Yes
Create support tickets	Yes
Add to wishlist	Yes
Access seller dashboard	Yes

Restrictions:

- Cannot purchase own listed cars
- Cannot list cars without admin approval
- Cannot modify other users' listings

4.2 Admin (ADMIN) Role

System Administrator: Controls moderation, approvals, and platform analytics.

Permissions:

Action	Allowed
Approve/reject car listings	Yes
Approve/reject purchase orders	Yes
Manage users (edit details)	Yes
View all orders and cars	Yes
Access analytics dashboard	Yes
View and respond to support tickets	Yes
View all user feedback	Yes

System Limitations:

- Only one ADMIN per system
 - Cannot transfer ADMIN role
 - Cannot delete records (soft delete only via rejection)
-

5. Buyer's Guide**5.1 Browsing Cars**

Endpoint: GET /api/cars/available

View Available Listings Buyers see all approved and available cars (excluding their own listings).

Each car listing includes:

```
{
  "id": 1,
  "make": "Honda",
  "model": "City",
  "year": 2021,
  "price": 850000,
  "mileage": 25000,
  "color": "White",
  "available": true,
  "condition": "GOOD",
  "fuelType": "PETROL",
  "transmission": "MANUAL",
  "bodyType": "SEDAN",
  "numberOfOwners": 1,
  "engineCc": 1497,
  "insured": true,
  "pucValid": true,
```

```

    "viewCount": 4,
    "seller": {
      "id": 2,
      "username": "seller1",
      "name": "Vikram Singh"
    },
    "createdAt": "2026-06-01T10:00:00"
  }
}

```

5.2 Car Details & Comparison

View Details: GET /api/cars/{carId}

- Each view increments the car's popularity counter
- Recorded in your “Recent Views” for future reference

Compare Cars: GET /api/cars/compare?firstCarId=1&secondCarId=2

- Returns detailed specs for side-by-side comparison
- Includes all condition fields (fuel type, transmission, etc.)
- Helpful for making purchase decisions

Popular Cars: GET /api/cars/popular

- Top 5 most viewed cars in the system
- Updated in real-time based on view counts

5.3 Making a Purchase

Endpoint: POST /api/user/cars/{carId}/purchase

Step 1: Initiate Purchase

```

{
  "paymentMethod": "UPI",
  "paymentToken": "pay_demo_token"
}

```

Payment Methods Supported: - UPI - Credit Card - Debit Card - Net Banking - Wallet

Step 2: Payment Processing

- Simulated payment gateway processes your transaction
- Use paymentToken: "fail_anything" to test payment failures
- Successful payment creates order in PENDING_ADMIN_APPROVAL state

Success Response:

```

{
  "id": 1,
  "buyer": {
    "id": 3,
    "username": "buyer1",
    "name": "Amit Kumar"
  },
  "car": {
    "id": 1,
    "make": "Honda",
    "model": "City"
  }
}

```

```

    },
    "payment": {
      "status": "SUCCESS",
      "amount": 850000,
      "method": "UPI"
    },
    "status": "PENDING_ADMIN_APPROVAL",
    "fraudAlert": false,
    "createdAt": "2026-06-06T21:00:00"
  }
}

```

Failure Response:

```

{
  "message": "Payment failed. Purchase flow cancelled. Payment ID: <paymentId>"
}

```

5.4 Purchase Workflow

The purchase follows a **two-step approval process**:

Step 1: Buyer Purchases

↓

Step 2: Admin Reviews & Approves (PENDING_ADMIN_APPROVAL)

↓

Step 3: Seller Reviews & Approves (PENDING_SELLER_APPROVAL)

↓

Step 4: Purchase Complete (APPROVED)

5.5 My Orders

Endpoint: GET /api/orders/my

View all orders where you are the buyer or seller:

```

[
  {
    "id": 1,
    "buyer": { "id": 3, "username": "buyer1" },
    "seller": { "id": 2, "username": "seller1" },
    "car": { "id": 1, "make": "Honda", "model": "City" },
    "status": "APPROVED",
    "createdAt": "2026-06-06T21:00:00"
  }
]

```

5.6 Download Receipt

Endpoint: GET /api/orders/{orderId}/receipt

Detailed receipt includes:

```

{
  "orderId": 1,
  "receiptNumber": "RCP-1-2026-06-06",
  "buyerName": "Amit Kumar",
  "buyerEmail": "buyer@example.com",
  "sellerName": "Vikram Singh",

```

```
"carMake": "Honda",
"carModel": "City",
"amount": 850000,
"paymentStatus": "SUCCESS",
"orderStatus": "APPROVED",
"generatedAt": "2026-06-10T09:45:00"
}
```

6. Seller's Guide

6.1 List Your Car

Endpoint: POST /api/cars

Complete Listing Form:

```
{
  "make": "Honda",
  "model": "City",
  "year": 2021,
  "price": 850000,
  "mileage": 25000,
  "color": "White",
  "condition": "GOOD",
  "fuelType": "PETROL",
  "transmission": "MANUAL",
  "bodyType": "SEDAN",
  "numberOfOwners": 1,
  "engineCc": 1497,
  "insured": true,
  "pucValid": true,
  "description": "Single owner, full service history, accident-free"
}
```

Condition Options:

- EXCELLENT - Like new condition
- GOOD - Well maintained
- FAIR - Some wear and tear
- POOR - Significant damage/repairs needed

Fuel Types:

- PETROL, DIESEL, ELECTRIC, HYBRID, CNG, LPG

Transmission Types:

- MANUAL, AUTOMATIC, CVT, AMT

Listing Status:

- New listings start as PENDING_ADMIN_APPROVAL
- Not visible to buyers until approved
- Cannot be edited until approval

6.2 Manage Listings

Endpoint: PUT /api/cars/{carId}

Update listing details:

```
{
  "price": 825000,
  "mileage": 26000,
  "description": "Updated service records available"
}
```

Note: Non-admin edits reset approval status to PENDING_ADMIN_APPROVAL

Delete Listing: DELETE /api/cars/{carId}

6.3 Seller Dashboard

Endpoint: GET /api/user/dashboard

View your business metrics:

```
{
  "totalListings": 10,
  "availableListings": 7,
  "soldListings": 3,
  "pendingListings": 2,
  "rejectedListings": 1,
  "pendingAdminOrders": 1,
  "pendingApprovals": 2,
  "approvedOrders": 3,
  "rejectedOrders": 1,
  "fraudAlerts": 0,
  "totalRevenue": 2500000
}
```

6.4 Pending Sales Approval

Endpoint: GET /api/user/sales/pending

View orders awaiting your approval:

```
[
  {
    "id": 5,
    "buyer": { "id": 3, "username": "buyer1" },
    "car": { "id": 1, "make": "Honda", "model": "City" },
    "payment": { "status": "SUCCESS", "amount": 850000 },
    "status": "PENDING_SELLER_APPROVAL"
  }
]
```

6.5 Approve/Reject Sales

Approve Sale: POST /api/user/sales/{orderId}/approve

```
{
  "message": "Sale approved",
  "status": "APPROVED"
}
```


- Car becomes permanently unavailable
- Purchase is finalized
- Buyer can download receipt

Reject Sale: POST /api/user/sales/{orderId}/reject

```
{
  "message": "Sale rejected",
  "status": "REJECTED"
}
```

- Car returns to available status
 - Buyer is notified
-

7. Admin Guide

7.1 Admin Dashboard

Endpoint: GET /api/admin/dashboard

High-level platform statistics:

```
{
  "totalUsers": 20,
  "admins": 1,
  "users": 19,
  "totalCars": 30,
  "availableCars": 22,
  "soldCars": 8,
  "pendingCarApprovals": 4,
  "rejectedCars": 1,
  "totalOrders": 15,
  "pendingAdminOrders": 4,
  "pendingOrders": 2,
  "approvedOrders": 8,
  "rejectedOrders": 2,
  "fraudAlerts": 0,
  "openTickets": 3,
  "closedTickets": 5,
  "totalRevenue": 6500000
}
```

7.2 Analytics & Charts

Endpoint: GET /api/admin/dashboard/graph

5 datasets for platform analytics (last 30 days):

```
{
  "ordersPerDay": [
    { "date": "2026-05-12", "count": 2 }
  ],
  "revenuePerDay": [
    { "date": "2026-05-12", "revenue": 1700000 }
  ],
  "carsListedPerDay": [
    { "date": "2026-05-12", "count": 3 }
  ]
}
```

```

    ],
    "orderStatusBreakdown": [
      { "status": "PENDING_ADMIN_APPROVAL", "count": 4 },
      { "status": "APPROVED", "count": 8 }
    ],
    "carApprovalBreakdown": [
      { "status": "PENDING_ADMIN_APPROVAL", "count": 4 },
      { "status": "APPROVED", "count": 20 }
    ]
  }
}

```

7.3 Car Moderation

Pending Listings: GET /api/admin/cars/pending

Approve Car: POST /api/admin/cars/{carId}/approve

```

{
  "approvalStatus": "APPROVED",
  "available": true
}

```

Reject Car: POST /api/admin/cars/{carId}/reject

```

{
  "approvalStatus": "REJECTED",
  "available": false
}

```

7.4 Order Management

All Orders: GET /api/admin/orders

Pending Orders: GET /api/admin/orders/pending

Approve Order: POST /api/admin/orders/{orderId}/approve

Moves to seller approval stage

Reject Order: POST /api/admin/orders/{orderId}/reject

Cancels purchase

7.5 User Management

List Users: GET /api/admin/users

Get User: GET /api/admin/users/{userId}

Edit User: PATCH /api/admin/users/{userId}

```

{
  "name": "Rahul Kumar",
  "phoneNumber": "9876543211",
  "email": "newemail@example.com",
  "newPassword": "newpassure123456"
}

```

8. Common Features

8.1 Wishlist

Add to Wishlist: POST /api/user/wishlist/{carId}

- Only approved, available cars allowed
- Duplicate additions are idempotent

View Wishlist: GET /api/user/wishlist

```
[
  {
    "id": 1,
    "car": { "id": 1, "make": "Honda", "model": "City" }
  }
]
```

Remove from Wishlist: DELETE /api/user/wishlist/{carId}

8.2 Seller Reviews

Submit Review: POST /api/reviews/sellers/{sellerId}

```
{
  "rating": 4,
  "comment": "Smooth transaction, car in great condition"
}
```

- Rating: 1-5 (required)
- Comment: Optional (max 1000 characters)
- Only eligible after APPROVED purchase
- Updates existing review (upsert)

View Reviews: GET /api/reviews/sellers/{sellerId}

Average Rating: GET /api/reviews/sellers/{sellerId}/average

```
{
  "sellerId": 2,
  "averageRating": 4.3
}
```

8.3 Recent Views

Endpoint: GET /api/cars/recent

Last 10 cars you viewed in current session:

```
[
  {
    "id": 5,
    "car": { "id": 1, "make": "Honda", "model": "City" },
    "viewedAt": "2026-06-06T21:00:00"
  }
]
```

9. Support & Feedback

9.1 Support Tickets

Create Ticket: POST /api/user/support-tickets

```
{
  "subject": "Payment issue",
  "description": "Money deducted but order not created"
}
```

My Tickets: GET /api/user/support-tickets

Add Message: POST /api/user/support-tickets/{ticketId}/messages

```
{
  "message": "Any update on this?"
}
```

Admin View Tickets: GET /api/admin/support-tickets

Admin Update Ticket: PATCH /api/admin/support-tickets/{ticketId}

```
{
  "status": "CLOSED",
  "response": "Issue has been resolved."
}
```

9.2 Feedback

Submit Feedback: POST /api/feedback

```
{
  "message": "Great platform, smooth experience"
}
```

Admin View Feedback: GET /api/feedback/admin

10. API Reference & Error Handling

10.1 HTTP Status Codes

Code	Meaning	Example
200	Success	Car retrieved
201	Created	Order created
400	Bad Request	Invalid input
401	Unauthorized	Missing session token
403	Forbidden	Not authorized for action
404	Not Found	Car ID doesn't exist
409	Conflict	Duplicate username
500	Server Error	Database error

10.2 Error Response Format

All errors return:

```
{
  "message": "Error description"
}
```

10.3 Common Validation Errors

Scenario	Error	Fix
Short password	“password must be at least 10 characters”	Use 10+ chars
Invalid phone	“phoneNumber must be 10 digits starting with 6-9”	Use valid Indian number
Duplicate username	“username already exists”	Choose unique name
Non-unique email	“email already in use”	Use different email
Buying own car	“Cannot purchase your own listing”	Purchase from other seller
Unapproved car	“Car is not approved”	Wait for admin approval
Invalid rating	“rating must be between 1 and 5”	Use 1-5 scale

10.4 Authentication Errors

Error	Cause	Solution
Missing X-Session-Token	No auth header	Add token to request
Invalid session token	Wrong/expired token	Login again
Session expired	Token older than 120 min	Login to get new token
Insufficient permissions	Role doesn't allow action	Use correct account type

10.5 Business Rule Violations

Rule	Error	Details
Car availability	“Car is not available”	Already sold
Admin uniqueness	“Only one admin allowed”	Admin exists
Seller reviews	“Not eligible to review”	No approved purchases
Fraud detection	“fraudAlert: true”	>3 total purchases

Quick Reference

Account Credentials (Demo)

Admin Account:

Username: demoadmin
Password: Demo@1234x

User Account:

Username: demouser
Password: Demo@1234

Key Endpoints Summary

Feature	Method	Endpoint
Register	POST	/api/auth/register
Login	POST	/api/auth/login
Browse Cars	GET	/api/cars/available
List Car	POST	/api/cars
Purchase	POST	/api/user/cars/{id}/purchase
Approve Order	POST	/api/admin/orders/{id}/approve
Approve Sale	POST	/api/user/sales/{id}/approve
My Orders	GET	/api/orders/my
Admin Dashboard	GET	/api/admin/dashboard
Support Ticket	POST	/api/user/support-tickets

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Contact: support@trustlot.com

For additional support, please create a support ticket through the application.